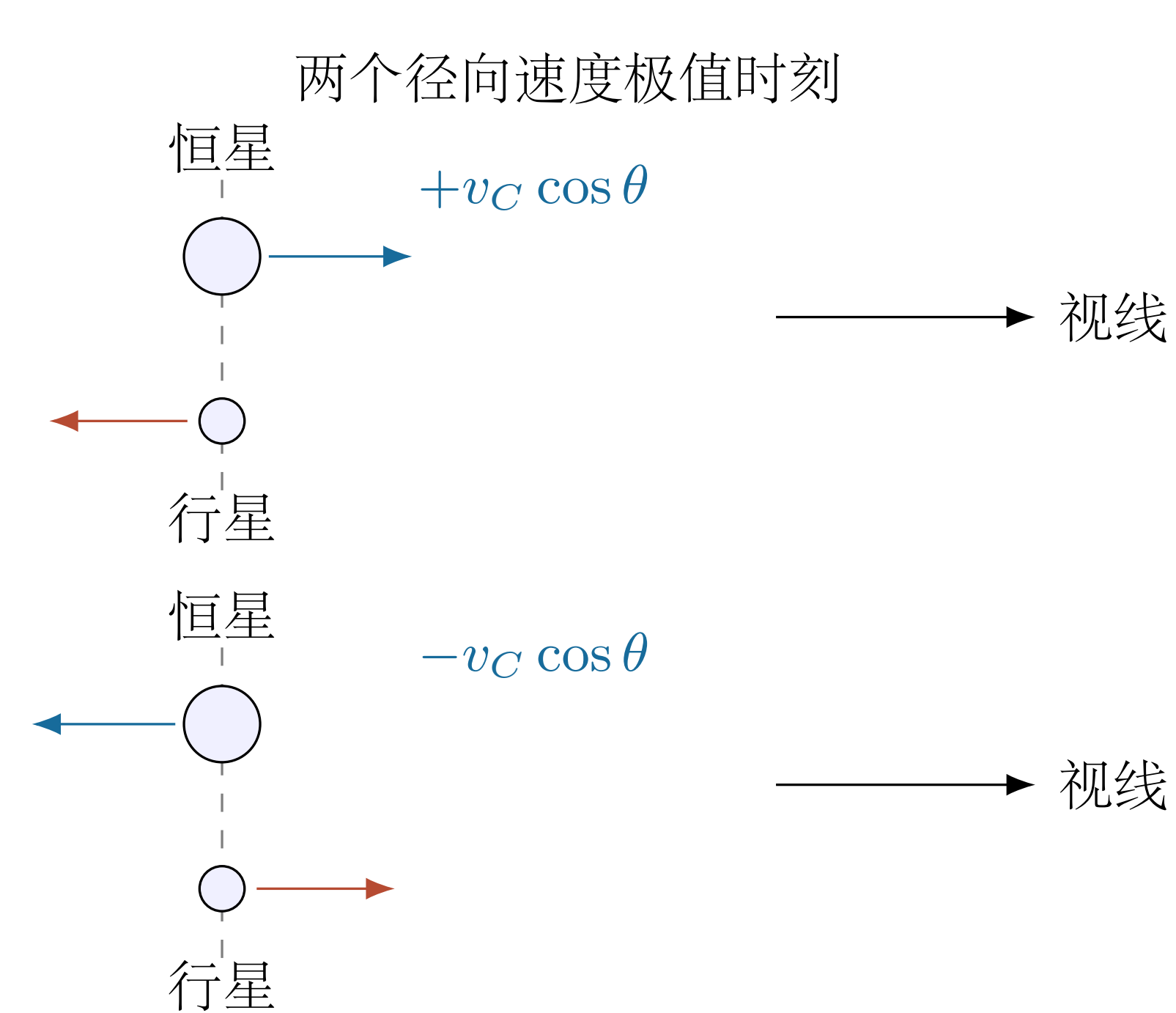

$$\Delta\nu/\nu = 2v_C\cos\theta/c$$